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Principles in Project Portfolio Management: Building Upon What We Know to Prepare for the Future

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Abstract

Since the 1950s, project portfolio management (PPM) has offered principles helping us to act in PPM-related situations. Our study integrates what we know by reviewing seven decades of PPM research. We identify 17 PPM principles in four categories that have shaped our thinking and show how these are embedded in a semantical hierarchy. By doing so we shed light on the thoughts, intentions, and practices behind the principles, which support practitioners in becoming knowledgeable agents. By reflecting on the deficits of the identified principles' underlying assumptions, we propose a path for future research.

Keywords

project portfolio management, principles, systematic literature review, emancipatory, engaged practice, engaged scholarship, ethics, sustainability

Introduction

Project portfolio management (PPM) has become an established research discipline during the last two decades (Hobbs, 2012). One important effort of the literature is to offer principles that help us to act in PPM-related situations. In that regard, principles can be understood as fundamental truths or propositions that serve as the foundation for a system of behavior or reasoning (Sankaran et al., 2021). Principles can be astonishingly resilient over time because they do not dictate what to do in specific situations. Instead, they more broadly tell us how to think in different situations and thus enable us to act in unknown circumstances (Bragg, 2018). Our current time is characterized by an increasing pace of change and more uncertainty (Teece et al., 2016). Thus, raising our awareness of what we know about PPM principles becomes more relevant than ever before as their guidelines help us cope with future and unknown challenges.

However, despite the importance of PPM principles, we find that the literature has not adequately acknowledged the principles' underlying thoughts, intentions, and assumptions. Such information is crucial in supporting practitioners in becoming knowledgeable agents. Moreover, we find no contributions addressing whether or how the discipline's existing principles will meet future demands. For instance, the future may demand an increase in collaborative ways of working, technological advances, more environmental issues, and more ethical responsibility (Walker & Lloyd-Walker, 2019). Thus, our article proposes a better understanding of PPM principles to support the evolution of the discipline as it adapts to meet the challenges of the future.

We argue that a systematic literature review is advantageous to address these questions as it would help us build on what we already know and point to the future (Denyer & Tranfield, 2009). We also emphasize the value of looking back at what has been written as important challenges for the next decades have already taken shape (Drucker et al., 1997). Such past-future thinking also sharpens our vision of the present—not only of the past—as it creates a baseline to evaluate the changes that will follow (Thomson, 2001).

Thus, our study traces the PPM literature back to its origin (i.e., Markowitz, 1952) by drawing on a database of 669 articles from seven decades of PPM research across multiple fields. We analyze a sample of influential articles and lay the groundwork for helping scholars and practitioners understand the intentions and assumptions underlying PPM principles. Revealing such assumptions invites reflection and conversation on the current take-it-for-granted state of the PPM discipline (Geraldi & Söderlund, 2018). It also puts us in an advantaged position to propose directions for the future (Baumeister & Leary, 1997). In sum, the aim of our research can be formulated with the

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questions: *(1) How can we understand the intentions and assumptions underlying PPM principles? (2) How can we lay out a path for the future?*

We organize our article as follows. The next section reviews what is currently known in the literature about PPM principles. The methodology section describes the steps taken in our structured literature review. The analysis section identifies PPM principles and shows how they can be understood as embedded in a semantical hierarchy. In the discussion section, we discuss how the knowledge of PPM principles can be advanced and suggest a future agenda. Finally, the last section presents our conclusions and the limitations of our research.

Principles and Project Portfolio Management

Principles have been widely used in PPM ever since the discipline originated with Nobel Prize winner H. Markowitz's famous mathematical evidence of diversifying principles to lower the overall risk in a portfolio (Markowitz, 1952). One of the ongoing challenges for PPM researchers who propose models based on such mathematical argumentation is demonstrating their usefulness in practice due to their complexity and abstractness (Gear, 1974). Often their recommendations require the support of a specialist to be usable by real-world organizations (Henriksen & Traynor, 1999). Since the 1960s, R&D literature and the new product development (NPD) literature have made excellent progress in developing principles for the practical application of PPM (Baker & Pound, 1964). In particular, the introduction of the Stage-Gate® model by Robert Cooper and his colleagues (Cooper, 1990; Cooper et al., 2000) made a large impact on practice. It offered the underlying principle that projects can be best managed by identifying a sequence of phases and ensuring that the project meets specific criteria before moving to the next phase. The Stage-Gate® model and the work by Cooper and colleagues still stand as some of the most influential PPM research.

During the last three decades, project management journals and books have gradually overtaken the NPD literature on developing PPM and its guidelines (Hansen & Svejvig, 2022; Hobbs, 2012). The project management associations have also played a significant role in developing principles via their standards such as the Project Management Institute (PMI, 2021), the International Project Management Association (IPMA, 2015), and the Office of Government Commerce (OGC, 2006). More recently, standards inspired by agile thinking and principles have emerged and have been widely used (Horlach et al., 2019; Knaster & Leffingwell, 2018; Leffingwell, 2007). The latest tendency is to adopt hybrids between agile and plan-driven as many agile implementations are, in reality, hybrid approaches (Gemino et al., 2021), where the organizational macro-planning level continues to use traditional practices (e.g., stage-gate models) (Cooper & Sommer, 2018).

As we close our historical overview of principles in the PPM literature, we notice that some approaches have gone out of favor over time. However, their underlying principles seem resilient. For instance, the mathematical approach to portfolio management often associated with Markowitz (1952) helped establish the principle of balancing the overall portfolio. This idea seems to endure over time and continues to be relevant for organizations in the future.

Defining Principles as Embedded in a Semantical Hierarchy

Principles can be considered a sophisticated construct of interrelated thoughts, intentions, interests, and practices (Gioia et al., 2013). In that regard, we find the term *semantical hierarchy* from Gemino et al. (2021) to be powerful in revealing how PPM principles are embedded in the order of linguistic elements. In the following, we show how the existing literature has laid the groundwork by offering terms such as waves, approaches, principles, and practices. We start with the most abstract term and move downwards, and we close the section by discussing how our research builds on what we already know.

Waves, as formulated by Lichtenberg (1989), can be used as an overarching term to define a paradigmatic shift in thinking behind the PPM literature. Other scholars have used the word generations to describe the development of models in R&D and NPD (Cooper, 1994; Nobelius, 2004; Rothwell, 1994; Roussel et al., 1991). The latter indicates a much more incremental development and does not point to fundamental shifts in thinking.

An *approach* can be used to describe how to govern projects. Inspired by Gemino et al. (2021), a PPM approach can be seen as a set of principles that provides strategic guidance on managing a portfolio of projects. Approaches include agile approaches (Knaster & Leffingwell, 2018; Leffingwell, 2007), traditional approaches (Archer & Ghasemzadeh, 1999), and hybrid approaches (Cooper & Sommer, 2016; Olsson et al., 2018).

Principles are largely nonprescriptive and thus differ from rules-based guidelines (Müller, 2016). They are abstract, but they can be converted into actions. This is primarily done through *practices*, which we define as techniques or procedures used to exercise aspects of one or more principles in project portfolios. Practices can also be combined to support a principle (Gemino et al., 2021).

As we close our discussion of what we currently know about the linguistic layers encapsulating PPM principles, we find potential for improvement in PPM research by creating a coherent and data-based overview. This leaves us with the unmet need of supporting practitioners to become more knowledgeable agents when using PPM of principles; thus, it offers us an opportunity to provide a contemporary contribution to the literature (Müller & Klein, 2018). The following section describes how our research is designed to pursue this goal.

Methodology

Drawing on Thomé et al.'s (2016) approach to structured literature reviews, we organized our study in six stages: (1) planning

and formulating the problem, (2) searching the literature, (3) establishing an initial data structure, (4) selecting a sample, (5) analyzing the data, (6) displaying the results. These stages are visualized in Figure 1, and the following text describes our procedures in each stage.

Stage 1: Planning and Formulating the Problem

First, we defined our research scope as integrating what we know about PPM principles. We aimed to review seven decades of PPM research and use this knowledge to draw paths for the future. We applied the narrative review method (Snyder, 2019), which, according to Klein and Müller (2020), acts as an intermediate between systematic reviews (e.g., Tranfield et al., 2003) and integrative reviews (Torraco, 2005, 2016).

Stage 2: Searching the Literature

In this stage, we utilized a prior literature review conducted by Hansen and Svejvig (2022) as this study offers a database of PPM articles covering the discipline since the early 1950s. The study above started with the use of an explorative search for previous literature reviews and conceptual papers on PPM. This resulted in 29 papers being analyzed, and backward and forward tracking was used to identify the main keywords in the discipline (Webster & Watson, 2002).

The application of structured search techniques has many pitfalls (Drake et al., 2021). One major challenge was that the keywords varied across the journals and the decades. Moreover, it was demanding to trace literature back to its original sources. In some cases, for almost 70 years, none of the available databases covered the entire period or all relevant journals. To get a satisfying result, we utilized six databases: JSTOR, Scopus, Science Direct, IEEE Xplore database, the *Project Management Journal*® database, and Google Scholar; specific search strings were defined for each database. The search was supplemented with the more flexible forward and backward tracking method to find relevant articles. The scope was limited to articles from peer-reviewed academic journals (Hansen & Svejvig, 2022).

Two criteria were used to assess the quality of the search so that our search did not artificially restrict the number of studies yet was specific enough to bring the studies related to the topic (Thomé et al., 2016). The search strings were refined in various iterations so that the final search result fulfilled the two quality criteria. The first criterion was to include 10 selected benchmark articles, and the second criterion was to cover the time span between 1950 and 2022 of the PPM discipline (see Appendix A). Our search resulted in a stock of 1,970 articles.

The next task was to filter the search result, and this process was done by reviewing the titles and abstracts of 1,970 articles. A formal reliability check of the process was completed, as two researchers conducted the filtering process until the criteria for inclusion were robust. Robustness was evaluated by comparing the coding results and discussing discrepancies until an agreement was reached. The inclusion/exclusion criteria were completely formalized after filtering approximately 1,000 articles, in other words, 50% of the filtering as advised by Thomé et al. (2016) (see Appendix B). Hereafter, one researcher was able to do the filtering from this point. Completing this process resulted in a stock of 669 articles covering seven decades of PPM research (Hansen & Svejvig, 2022), and this stock will be utilized in this article.

Stage 3: Establishing an Initial Data Structure

In this stage, we used the theoretical framework with the four waves originally proposed by Lichtenberg (1989) and further developed by Sankaran et al. (2021). We used the framework to create a basis to inform our study, make an initial structure for later analysis, and divide data into smaller and more feasible chunks of data. Table 1 displays our codebook, outlines our definition of the four waves, gives an example of an article from each wave, and shows the number of articles that we associated with each wave.

We started the coding process by reading the abstracts of the 669 articles, and we coded the articles with the publication year and the wave we found to be the most predominant. As most articles touched on more than one wave, we found it most accurate to code each contribution as attached to more than one category (Drake et al., 2021). To ensure quality, we reviewed the codes at two different points in time (Hiebl, 2021)—December 2021 and February 2022—and found that 3% of the codes needed to be revised. We displayed and assessed the coding results and found the four waves valuable to establish an initial structure for our analysis. The abstract nature of the theory gave us a high degree of freedom to modify and develop its assumptions (Walsham, 1995).

Stage 4: Identifying a Sample

A total of 669 articles would be a very high number to analyze compared to similar literature reviews (Hiebl, 2021). Hence, we wanted to select a sample with a comprehensive representation of data. This does not equate with including "all research items" but reaching *data saturation*, which we define as containing enough information to replicate the study (O'Reilly & Parker, 2013) and when adding more data not providing new knowledge (Fusch & Ness, 2015). On this backdrop, we selected the 25 most cited articles in each wave by looking at Google Scholar (GS citations), which, despite reported shortcomings (Pozsgai et al., 2021), is appropriate for evaluating impact in practical-orientated disciplines (Chapman & Ellinger, 2019). Using GS citations provided a data-driven and transparent process in our sample selection. We found that the effect of adding more articles to the analysis became insignificant. Thus, this indicated that the sample of 100 articles

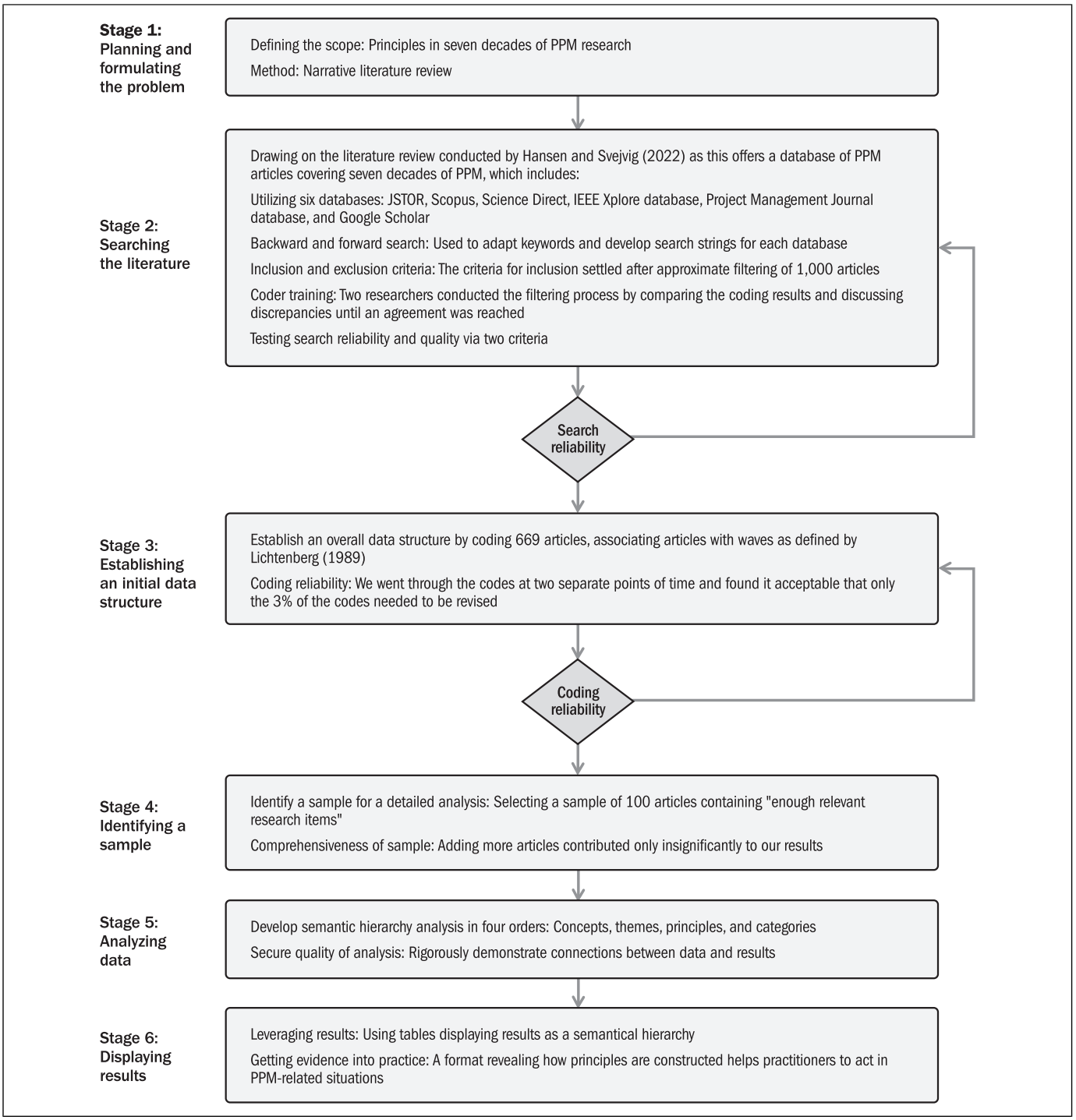

Figure 1. Six stages in our literature review process.

(the 25 most cited articles from each of the four waves) provides enough data to fulfill our saturation criteria.

Stage 5: Analyzing Data

In this stage, we identified PPM principles and the semantic hierarchy constituting these principles. Our purpose was to lay the groundwork for helping scholars and practitioners better understand PPM principles. Broadly inspired by Gioia et al. (2013), we organized our analysis in different orders. In the first-order analysis, we identified the *concepts*. This was done by sequentially going through the selected articles from the four waves. In the second-order analysis, we developed the identified concepts into *themes*. This led to the third-order

Table 1. Defining and Exemplifying the Four Waves. Adapted from Lichtenberg (1989) and Sankaran et al. (2021)

Wave (code)	Definition	Number of Articles Associated with Each Wave	Example
First wave	Involves formalization and a focus on tools and techniques	467	Hess (1962) proposes a dynamic programming approach to R&D budgeting and project selection.
Second wave	Is about logical network-based approaches	177	Schmidt and Freeland (1992) include the organizational context in which decisions are made by considering decisions as a part of the system.
Third wave	Has a "humanized" focus and use of tools to serve overall goals	138	Blichfeldt and Eskerod (2008) address the dilemma of wanting to include all projects in PPM and aiming to keep the burden of doing PPM reasonably.
Fourth wave	Sees project portfolios as temporary phenomena and paradoxes	41	Petro (2019) proposes ambidexterity to pursue projects with contradictory strategic goals simultaneously.

analysis in which we gathered the themes into *principles*. Finally, at the most abstract level—our fourth-order analysis—we organized the principles into *categories*.

Stage 6: Displaying Results

We displayed the results of our fourth-order analysis in text and tables. We used the categories as the sections' subtitles and used tables to show the allied principles, themes, and concepts. The leftmost column in each table contains 25 article references—and each row represents a concept from one or more articles. The tables' central and rightmost columns show how concepts are merged into themes and principles. Such a visual display of the data structure makes it easier for practitioners and academics to engage in our results (Van de Ven, 2007). It also secures rigor and transparency in qualitative analysis (Gioia et al., 2013). The tables displaying the data structure can be found in Tables 2 through 5 in the analysis section.

A Semantical Hierarchy Encapsulating PPM Principles

The following analysis gives a detailed account of our semantical hierarchy, which at the most abstract level consists of the four categories: balance the portfolio, build a coherent PPM system, develop strategy effectiveness and stakeholder involvement, and create readiness to change and deal with paradoxes.

Balance the Portfolio

This category of principles is centered around the project selection process and, as shown in Table 2, we find it to contain five principles. These principles are: (P1) set priority criteria, (P2) formalize selection, (P3) use tools to support decision-making, (P4) deal with multiple project challenges, and (P5) comprehend the deficiencies of computed methods.

(P1) Set priority criteria: This principle is grounded in a stream of literature on multicriteria selection methods (Danesh et al., 2018), where the analytic hierarchy process (AHP) is very influential. It can help project-based organizations map their prioritization criteria and evaluate the relative importance of the criteria. In short, this is done by systematically comparing the criteria pairwise, and by using a simple method, the criterion's importance can be calculated. Using pairwise comparisons minimizes the cognitive load of dealing with many criteria. Thus, this method can help organizations in complex environments prioritize their projects to reflect their preferences (Liberatore, 1987; Meade & Presley, 2002).

(P2) Formalize selection: This principle offers guidelines on picking projects using formalized methods. The literature in our stock suggests using computed models and provides a variety of sophisticated mathematical programming. Mathematical programming is about finding the equation that provides the optimal trade-off between constraints and a defined goal. We found that our data utilized many types of programming, such as linear programming (Stummer & Heidenberger, 2003) nonlinear programming (Doerner et al., 2004), and dynamic programming (Weingartner, 1966).

(P3) Use tools to support decision-making: The use of matrixes to understand and visualize portfolios has been well known for many years (Cooper et al., 1999) and reported to have a positive impact on decision-making (Killen et al., 2020). In our stock of PPM literature, Mikkola (2001) suggested using a portfolio matrix to show the competitive advantages of firm projects. Dickinson et al. (2001) showed how a matrix could be used to quantify project interdependencies and thus more easily find the optimal selection of projects.

(P4) Deal with multiple project challenges: This principle looks at problems believed to be present in many organizations. Elonen and Artto (2003) emphasize (1) inadequate project-level activities; (2) a lack of resources, competencies, and methods; (3) a lack of commitment, unclear roles, and responsibilities; (4) inadequate portfolio-level activities; (5) inadequate information management; and (6) inadequate management of the

Table 2. Balance the Portfolio: The Category's Principles, Themes, and Concepts

Concepts	Themes	Principles
Use the analytic hierarchy process (AHP) for priority setting and resource allocation in the industrial R&D environment (Liberatore, 1987).	Priority setting using multicriteria approaches	(P1) Set priority criteria
Use the analytic network process (ANP), a general form of Saaty's AHP, as a model to evaluate the value of competing R&D project proposals (Meade & Presley, 2002).		
AHP and the analytic network process are applied to help the managing board to decide whether to invest and prioritize (Aragonés-Beltrán et al., 2014).		
Use a multicriteria approach to project complexity evaluation (Vidal et al., 2011).		
Handle project interrelationships such as mutual exclusion and interdependencies. The techniques utilized are linear and integer programming, dynamic programming, and the discrete optimizing procedure (Weingartner, 1966).	Computed models (miscellaneous)	(P2) Formalize selection
Use Pareto Ant Colony Optimization (Doerner et al., 2004).		
Rank project alternatives based on the criteria of relevance, risk, reasonableness, and return (Henriksen & Traynor, 1999).		
Identify project proposals and use a multiobjective integer linear programming model to determine the solution space (Stummer & Heidenberger, 2003).		
Use robust portfolio modeling and project selection to account for project interdependencies, incomplete cost information, and variable budget levels (Liesiö et al., 2008).		
Address the static resource-constrained multiproject scheduling problem with project lateness and portfolio lateness (Browning & Yassine, 2010).		
Use the real options valuation model (Wang & Hwang, 2007).	Computed models (real option)	
Use the real options model to incorporate the impact of project interdependencies to calculate the option value of all projects (Bardhan et al., 2004).		
Use the R&D project portfolio matrix to link the portfolio of R&D projects (1) to the competitive advantages of the firm and (2) to the benefits the customers will receive (Mikkola, 2001).	Portfolio matrix to support decisions	(P3) Use tools to support decision-making
Use a dependency matrix to quantify the interdependencies between projects. In addition, use a nonlinear, integer program model to optimize project selection (Dickinson et al., 2001).		
The need to manage project interdependencies assuring their mutual compatibility at the portfolio level (De Maio et al., 1994).		
Use a decision support system called Project Analysis and Selection System (PASS) (Ghasemzadeh & Archer, 2000).	Software-supported decisions	
Pay attention to known problem areas in PPM: (1) inadequate project-level activities; (2) lacking resources, competencies, and methods; (3) lacking commitment, unclear roles, and responsibilities; (4) inadequate portfolio-level activities; (5) inadequate information management; and (6) inadequate management of the project-oriented organization (Elonen & Artto, 2003).	Problems in multiple projects environments	(P4) Deal with multiple project challenges
The importance of single-project management in PPM (Martinsuo & Lehtonen, 2007).	Portfolio management efficiency	
Assess the types of risks in the projects and adjust the project management approach accordingly (McFarlan, 1981).	Contingency model to adjust project management	
Understand the resource allocation syndrome as a more profound organizational problem (Engwall & Jerbrant, 2003).	Resource allocation problems	
Understand the limitations of existing normative R&D benefit measurement and project selection models (Baker, 1974).	The shortcomings of mathematical methods	(P5) Comprehend the deficiencies of computed methods
Understand the shortcomings of selection models. They are not		

(continued)

Table 2. (continued)

Concepts	Themes	Principles
tested empirically and are not feasible in practice (Baker & Pound, 1964).		
Use models with idea submittal arrangements and soft benefit evaluation to meet the limitations of financial models (Cooper et al., 2001; Hall & Nauda, 1990; Kennedy, 1997).	Less mathematically complex selection and evaluation model	

project-oriented organization. Other scholars have focused entirely on resource allocation problems, which are profound and enduring problems in multiproject environments (Engwall & Jerbrant, 2003).

(P5) Comprehend the deficiencies of computed methods: Scholars have for many years reported how uncertainty has undermined the usefulness of computed models. Studies have shown that it is often impossible to predict the most beneficial projects. For example, Klein and Meckling's (1958) studies of R&D development during the Second World War showed that an ongoing and skeptical approach is superior to mathematical calculations made early in the selection process.

Build a Coherent PPM System

We define *a coherent PPM system* as a unified whole with exchangeable components across its entities. This category expands the former's narrow focus on tools and techniques by including processes and governance, making PPM an integrated system. This distinction has many similarities with Schmidt and Freeland's (1992) distinction between the *decision events* and *system approaches*. As shown in Table 3, we identify four principles in this category: (P6) increase maturity by adopting best practices, (P7) apply a process view, (P8) develop governance, and (P9) create a supportive culture.

(P6) Increase maturity by adopting best practices: This principle has been particularly influential since the millennium, as the literature suggests maturity models (De Reyck et al., 2005; Jeffery & Leliveld, 2004). These models assume that adopting a set of practices is associated with superior outcomes (or improved outcomes), for example, by improving portfolio performance, providing more value, or decreasing project failures. However, these best practices usually come from correlational research with inconclusive empirical evidence (statistical significance and response rate in the inquiries) (Hansen & Kræmmergaard, 2014).

(P7) Apply a process view: Driven by the literature on stage-gate models, this principle introduces the notion that PPM is a process that supports projects from idea to launch and beyond. One key idea suggested by stage-gate models is to build effective go/kill decision points. This means that projects can be stopped if they become less valuable than expected or if other projects are a better way to use organizational resources. Whereas stage-gate models take a process view on PPM, other models extend this scope by applying a more cohesive view with a comprehensive set of practices supporting each step (e.g., Archer & Ghasemzadeh, 1999).

(P8) Develop governance mechanisms: This principle is related to the former but differs by offering concepts to understand how PPM-related elements should be vertically and horizontally connected. Müller et al. (2008) described control mechanisms to link project activities and management vertically. Varma et al. (2007) provided an example of horizontal coordination using a network model to understand how cross-functional coordination can improve enterprise performance and growth prospects.

(P9) Create a supportive culture: The business's climate and culture for innovation seem to be one of the strongest drivers of PPM performance (Cooper et al., 2004a). For example, Cooper et al. (1999) showed that an organization's PPM approach should meet management expectations in two ways, that is *management fit* (does it fit with the current senior management decision-making style) and *overall quality* rating (does management consider PPM to be efficient and thus not a waste of time). It seems difficult to overestimate the importance of having a culture with management support and buy-in, as this has been repeated by various studies (Cooper et al., 2000; Killen & Hunt, 2013). We will elaborate further on this topic in principle (P12), which is about stakeholder involvement.

Secure Strategy Effectiveness and Stakeholder Involvement

This category alludes to how PPM needs to move from a problem-focused to an organizational perspective using a strategic/business view. It also focuses on concepts such as personal issues and engaging, involving, and clarifying employees' roles at all organizational levels. As shown in Table 4, we find four principles in this dimension: (P10) focus on strategic effectiveness; (P11) apply a holistic value approach; (P12) focus on stakeholder involvement, roles, and responsibilities; and (P13) use alternative understandings.

(P10) Focus on strategic effectiveness: Effectiveness can be defined as the degree to which the portfolio successfully produces the correct and expected outcome (Patanakul & Milosevic, 2009). One helpful concept to evaluate whether results are preferred is the term alignment, and Srivannaboon (2006) suggested a quantified method to assess to which degree projects are aligned with the strategy. Effectiveness also implies some degree of efficiency in the desired

Table 3. Build a Coherent PPM System: The Category's Principles, Dimensions, Themes, and Concepts

Concepts	Themes	Principles
Raise maturity by improving attitude, knowledge, and actions in projects, programs, and the portfolio (Andersen & Jessen, 2003).	Maturity models	(P6) Increase maturity by adopting best practices
Understand maturity in four stages (ad hoc, defined, managed, and synchronized). Move up in maturity level to increase portfolio success (Jeffery & Leliveld, 2004).		
Assume that increasing adoption of best practices results in a reduction in project-related problems and better portfolio performance (De Reyck et al., 2005).		
Use best practices for PPM and assume a positive influence on performance (Killen et al., 2008).	Guided by benchmarks	
Use stage-gate methods to overview the entire new product process for senior managers. It provides a structure for better management and control (Cooper, 1990).	Stage-gate models	(P7) Apply a process view
Use the stage-gate process to move projects from idea to launch and beyond. It provides a blueprint for better management (Cooper, 2008; Cooper et al., 2004a, 2004b).		
Build more effective go/kill decision points. Gates are a two-part decision process: (1) to make go/kill decisions on individual projects; (2) to prioritize the project against the rest and allocate resources (Cooper et al., 1997a; Cooper et al., 2002).		
Use stage-gate models and integrate the gating system in PPM (Cooper et al., 2000)		
Apply a process view on the front-end activities (Khurana & Rosenthal, 1998).	Front-end activities	
Harness and reproduce technological capabilities by integrating knowledge, information flows, and actors (Gann & Salter, 2000).	An integrated model	
Separate the PPM system into distinct stages. Each stage accomplishes a particular objective and creates inputs for the next stage (Archer & Ghasemzadeh, 1999).		
Involve project teams, department members, and the portfolio management team in the planning processes (Platje et al., 1994).	Integrated planning process	
Implement four elements: (1) focus on selecting the right projects, (2) project sponsorship with a direct link between the executive and the project managers, (3) a project management office, and (4) project support securing an effective governance system (Too & Weaver, 2014).	Implement and link elements	(P8) Develop governance mechanisms
Intimately connect strategy and resource allocation (Cooper et al., 1997b).		
Recognize the importance of enterprise-wide, cross-functional coordination as it improves enterprise performance (Varma et al., 2007).	Horizontal coordination	
Recognize the importance of portfolio control mechanisms (Hansen & Kræmmergaard, 2013; Müller et al., 2008).	Portfolio control mechanisms	
Recognize the benefits of managing portfolios by using program frameworks (Ferns, 1991).	Program frameworks	
Adapt the organization's PPM approach to management's expectations and perceptions (Cooper et al., 1999).	Contingency model	(P9) Create a supportive culture
Consider culture of innovation within the business to be one of the strongest drivers of NPD performance (Cooper et al., 2004a, 2004b).	Learn from the best performers	
Understand PPM decision-making as having different decision-making styles (evidence-, power-, and opinion-based), and each decision style demands distinct input (Kester et al., 2011).	Adapt decision-making style to the needs of the organization	

outcome. Value (or benefit) can be seen as a way to operationalize this outcome (Serra & Kunc, 2015).

(P11) Apply a holistic value approach: This dimension emphasizes that the notion of value may include noncommercial aspects—ecological, societal, and learning values (Martinsuo & Killen, 2014). Brook and Pagnanelli (2014) suggested integrating this endeavor into how organizations evaluate their projects' value, which may address future market, social, and environmental needs. Martinsuo and Killen (2014) echoed this call, stating that noncommercial aspects are not

Table 4. Strategy Effectiveness and Stakeholder Involvement: The Category's Principles, Dimensions, Themes, and Concepts

Concepts	Themes	Principles
Align project management with business strategy (Srivannaboon, 2006).	Implementing strategy through projects	(P10) Focus on strategic effectiveness
Manage strategic intentions through multiple projects. Draw on reflections from an empirical study (Dietrich & Lehtonen, 2005).		
Tailor product development to strategy. Draw the findings from the case of a European technology manufacturer (Loch, 2000).		
Apply a strategic orientation. This could be analytical posture, risk-taking posture, or aggressive posture (Meskendahl, 2010).		
Implement IS strategy through projects. Draw on evidence from a leader in new public management (Young et al., 2012).		
Use benefits realization management (BRM) to close the gap between strategy planning and execution (Serra & Kunc, 2015).	Value as an integrating means between strategy planning and execution	
Consider projects and programs as value-creation processes, and let value imbue the PPM discipline (Winter & Szczepanek, 2008).		
Consider effectiveness in managing a group of multiple projects (Patanakul & Milosevic, 2009).	Effectiveness	
Apply four recommendations: (1) integrate benefits and cost management into a holistic value management approach; (2) see value creation and value capture as a broad concept; (3) let value imbue the PPM discipline; (4) develop new models and theories by applying frameworks of independent theory, e.g., the resource-based view (Laursen & Svejvig, 2016).	Broadening the scope of value	(P11) Apply a holistic value approach
Identify and assess strategic, noncommercial value in project portfolios. Map the relevant dimensions of strategic value and supplement previous frameworks with the noncommercial aspects: ecological, societal, and learning values (Martinsuo & Killen, 2014).	Strategic, noncommercial value	
Integrate sustainability into innovative PPM (Brook & Pagnanelli, 2014).	Sustainability	
Manage stakeholder behavior (engagement intensity of senior management, line managers, portfolio managers, and project managers) (Beringer et al., 2012). The effects of the engagement intensity of senior managers on success are inverted u-shaped (Beringer et al., 2013).	Senior management involvement	(P12) Focus on stakeholder involvement, roles, and responsibilities
Enforce strategic fit of project portfolios by project termination using the term project termination quality (Unger, Kock, et al., 2012).		
Understand the practices, roles, and responsibilities of middle managers in program and portfolio management (Blomquist & Müller, 2006).	Portfolio and middle management involvement	
Know that the transformational leadership behavior of portfolio managers can have a positive and significant impact on project performance (Kissi et al., 2013).		
Empower project portfolio managers and understand how management involvement impacts PPM performance (Jonas, 2010).		
Understand the three roles of a PPM office and these roles' impacts on portfolio management execution and success (Unger, Gemünden, et al., 2012).		
Understand the importance of customer integration into PPM and use the term relationship value (Voss, 2012).	Customer relations	
Understand the importance of relationship value (*to* customers and *from* customers) (Voss & Kock, 2013).		
Decide whether or not all projects should be part of PPM. Including all projects is not feasible and may impact employer motivation (Blichfeldt & Eskerod, 2008).	PPM is more than rational decisions	(P13) Use alternative understandings

(continued)

Table 4. (continued)

Concepts	Themes	Principles
Consider the balance between order and chaos in multiproject firms. Use a conceptual model (Geraldi, 2008).		
Understand PPM using metaphors and analyzing basic assumptions (Eskerod, 1996).		
Avoid the vicious circle of planning and control (Platje & Seidel, 1993).		
Expand the levels of analysis and the types of inquiry when reflecting on PPM (Geraldi & Söderlund, 2018).	Broaden project and portfolio scope	

sufficiently accounted for in project portfolio frameworks and decision makers' collective sensemaking.

(P12) Focus on stakeholder involvement, roles, and responsibilities: This principle deals with stakeholders at all levels. Beringer et al. (2012) described senior management involvement using the concept of *intensity*. These scholars and their colleagues found intense senior management involvement may negatively impact performance, particularly in immature environments (Unger, Kock, et al., 2012). In addition, the results of portfolio managers' efforts are mixed and they did not find a significantly positive effect from the portfolio managers' work (Beringer et al., 2013). Voss (2012) and Voss and Kock (2013) directed our gaze to stakeholders outside the focal organization and showed the importance of customer relationships.

(P13) Use alternative understandings: This principle differs from prior principles in two ways. First, it takes a more empirically driven approach to how findings arise by letting findings result from a negotiation between theory and the empirical settings. For example, Blichfeldt and Eskerod (2008) used sensemaking theory as a flexible frame to analyze how project managers need degrees of freedom to stay motivated. Second, this principle breaks with the assumption in prior research that PPM is a state of unity and order (Geraldi, 2008).

Create Readiness to Change and Deal with Paradoxes

This category sees project portfolios as temporary phenomena with legacies that transform organizations and societies. While the means of this category are still emerging (Sankaran et al., 2021; Walker & Lloyd-Walker, 2019), we find that it contains four principles, as shown in Table 5. The principles are (P14) cope with uncertainty, (P15) establish preconditions for innovation and change, (P16) enable agility outside of individual projects, and (P17) deal with paradoxes in transitional times.

(P14) Cope with uncertainty: This dimension describes ways to develop capabilities enabling organizations to cope with fast-changing surroundings such as hypercompetitive environments (Biedenbach & Müller, 2012). Many publications have suggested that organizations develop dynamic capabilities (DC) (Feiler & Teece, 2014; Petit, 2012). DC can be analyzed by looking at processes, positions, and paths (Killen & Hunt, 2010; Teece et al., 1997) and developed through sensing, seizing, and transforming (Killen et al., 2012). Other articles have emphasized a dual focus on the external and internal environment and the institutional level to create the conditions to support and foster projects (Morris & Geraldi, 2011).

(P15) Establish preconditions for innovation and change: Midler (2013) and Maniak and Midler (2014) propose the concept of project-lineage management, which focuses on the fundamental processes driving exploration efforts and shaping innovation trajectories. Project-lineage management is a multiproject sequence beginning with a breakthrough project that introduces a new product concept, including subsequent projects that both build on and transform the initial concept. According to this principle, innovation is not an island but mostly depends on the path of previous projects across a portfolio over time (Maniak & Midler, 2014).

(P16) Enable agility outside of individual projects: Agile project management approaches caused a silent revolution in how projects are organized and executed. However, many agile approaches rely on conditions with small, colocated software projects and teams. To leave this comfort zone and implement agile approaches in a broader organizational context, research calls for a view of agility outside of individual projects (Stettina & Hörz, 2015). Sweetman and Conboy (2018) suggest accomplishing portfolio level agility by organizing the portfolio as a complex adaptive system where projects act as autonomous agents. Cooper (2016) proposes a less radical solution in the form of a hybrid between traditional and agile approaches. Cooper's article demonstrates how agile approaches with their new tools, such as sprints, scrums, burndown charts, and backlogs, can be integrated with the traditional stage-gate processes in PPM. The next principle (P17) discusses in more detail the paradoxes of adopting such hybrids.

(P17) Deal with paradoxes in transitional times: Eriksson (2013) highlighted the contradiction between a short-term focus on efficiency and exploiting existing knowledge and technologies and a long-term focus on innovation and strategic development based on exploring new knowledge and technologies. As both elements are vital for sustainable competitive advantage, companies need to manage both simultaneously in

Table 5. Create Readiness to Change and Deal with Paradoxes: The Category's Principles, Dimensions, Themes, and Concepts

Concepts	Themes	Principles
Apply a strategical orientation via resource-based view, dynamic capabilities, and absorptive capacity (Killen et al., 2012).	Dynamic capabilities	(P14) Cope with uncertainty
Use sensing, seizing, and transforming (Petit, 2012).		
Emphasize intrapreneurship, proactive adaptability, and strategic renewal (Sicotte et al., 2014).		
Identify and prioritize dynamic capabilities through a strategic assessment. This is built and evolved through dialogical, collaborative, and iterative processes informed by learning (Feiler & Teece, 2014).		
Focus on dynamic capabilities consisting of people, structures, and processes. These should be monitored and adjusted to meet the changing requirements of the dynamic (Killen & Hunt, 2010).		
Distinguish among first-order dynamic capabilities, second-order dynamic capabilities, and ordinary capabilities (Daniel et al., 2014).		
Develop maturity for the staged implementation and evolution of a robust and dynamic PPM capability (Killen & Hunt, 2013).		
Build operational capabilities at the project level and dynamic capabilities at the portfolio level. The latter appears to be the generative mechanism for top managers' influence over firm performance (Hermano & Martín-Cruz, 2016).		
Strategical orientation via dynamic capabilities, resource-based view, and absorptive capacity (Killen et al., 2012).		
Develop these capabilities: innovative, adaptive, and absorptive (Biedenbach & Müller, 2012).	External and internal environment-oriented approaches	
View PPM as negotiation and bargaining and structural reconfiguration, besides rational decision processes, responding to uncertainties and complexities in business environments. These alternative perspectives offer new insight into the dilemmas identified in PPM (Martinsuo, 2013).		
Focus on velocity, turbulence, growth, and instability (Floricel & Ibanescu, 2008).		
Focus on the institutional level, where management creates the conditions to support and foster projects both in its parent organization and its external environment (Morris & Geraldi, 2011).		
Focus on formality and explicitness, review frequency, decision transparency, and information support (Urhahn & Spieth, 2014).	Preconditions for innovation portfolios	(P15) Establish preconditions for innovation and change
To foster response to change: Five structural and cultural components are derived: (1) clarity of strategic goals, (2) formality of the IPM processes, and (3) controlling intensity serve a coordinating function. The cultural components of (4) innovation climate and (5) risk climate serve a motivating function in IPM (Kock & Gemünden, 2016).		
Focus on project-lineage management (Midler, 2013).	Lineage management	
Focus on the key processes that drive exploration efforts and shape innovation trajectory; this is defined as multiproject lineage management (Maniak & Midler, 2014).		
Focus on commitment, transparency, collaboration, and team orientation (Stettina & Hörz, 2015).	Agile at scale	(P16) Enable agility outside of individual projects
Integrate elements of the agile IT product development method into traditional gating (stage gates) (Cooper, 2016).	Agile stage-gate models	
Apply structural and sequential separation of exploration and exploitation (Eriksson, 2013).	Ambidexterity	(P17) Deal with paradoxes in transitional times
Develop knowledge management capabilities combined with capabilities to obtain greater ambidexterity performance (Bresciani et al., 2018).		
Use improvisation: Improvisation is deliberate, i.e., it results from		

(continued)

Table 5. (continued)

Concepts	Themes	Principles
intentional efforts and occurs during the actions (Jerbrant & Karrbom Gustavsson, 2013).	Balancing flexibility and structure by improvising	
Allow competitors to develop innovation capabilities through close resource and knowledge sharing (Fernandez et al., 2018).	Project structures to achieve competitive innovation projects	
Focus on the capability paradox: tensions between value creation and value protection (Srivastava & Gnyawali, 2011).	Capability paradox	
Apply strategic control of deliberate strategy and emerging strategy (by management) by using control mechanisms (Kopmann et al., 2017).	Fostering both deliberate and emergent strategy	

their project portfolios. Thus, it is suggested that organizations become ambidextrous, and three types of ambidextrousness are identified: (1) structural ambidexterity, which requires separating exploration and exploitation activities (e.g., Fernandez et al., 2018); (2) sequential ambidexterity, which involves dividing activities into a sequence; and (3) contextual ambidexterity, which means having the capability to simultaneously and synchronously pursue exploration and exploitation within a business unit (Eriksson, 2013). The tension and paradoxes in competitive environments are many. For example, organizations may foster deliberate and emergent strategies in parallel for better portfolio performance (Kopmann et al., 2017).

Discussion

We have presented our semantical hierarchy from which we have extracted 17 principles. Identification of these principles from the research literature has the potential to guide organizations in evaluating and improving PPM practices. Thus, the principles can be used as a reference point to reflect on how organizations should act in PPM-related situations. However, as we enter an era where ethical and environmental responsibilities are expected to escalate (Walker & Lloyd-Walker, 2019), we ask if the current principles are sufficient to meet future challenges.

One way to invite reflection and conversion to the current state is to look at the knowledge perspective underpinning the principles (Geraldi & Söderlund, 2018). Broadly inspired by Svejvig and Schlichter (2020) and Geraldi and Söderlund (2018), we divide PPM principles into three knowledge perspectives: technical, understanding, and emancipatory. Table 6 shows the three perspectives represented in the 17 principles (P1–P17). This leads us to debate what implications this has for the PPM discipline, and we use the three perspectives to structure our discussion.

Technical perspective refers to the ambition to control and predict the environment, using natural science as an ideal. We find this perspective dominant in the two categories, *balancing the portfolio* and *building a coherent PPM system*. The two categories are primarily based on positivistic research and aim to reveal regularities and "truth" and consider rational reasoning as an ideal (See P1–P9). Indeed, principles based on this perspective are influential in guiding practice. They serve as the means to reach key PPM goals such as (P1) finding priority criteria or (P2) selecting the right projects. Thus, the technical perspective constitutes a core of the PPM discipline, helping organizations fulfill their raison d'être, which is to create products. Notwithstanding, this perspective has shortcomings if used in isolation as it may reduce practitioners to trained technicians (Geraldi & Söderlund, 2016). Such a narrow conceptualization of PPM principles combined with nonreflective endeavors may also lead organizations to dubiously assume that their advice fits all contexts (Morris, 2013).

The *understanding perspective* category is driven by the desire to interpret and understand social interactions. This perspective is predominant in the category *secure strategy effectiveness and stakeholder involvement*, for instance in (P13), which suggests using alternative understandings. It is rooted in the human need to communicate with others and develop an intersubjective understanding of their context and relationships (Geraldi & Söderlund, 2018). In contrast to the two former categories, it considers PPM as phenomena or contexts to be understood. It starts by "zooming in" on the empirical settings and then "zooming out" (Nicolini, 2009, p. 3) by using theory to formulate advice based on the inquiry. This connects peoples' actions to the actuality of PPM and empowers the practitioners to become reflective and apply a broader conceptualization than the technical perspective (Svejvig & Schlichter, 2020). However, in isolation, the *understanding perspective* category tends to become too descriptive with relatively weak theorizing and implications that do not go beyond the focal organization's PPM.

The *emancipatory perspective* category occurs when critical reflection culminates in deliberate changes in the status quo (Habermas, 1972). Geraldi and Söderlund (2018) expand Habermas's definition of emancipation and used the term as an opportunity to introduce new concepts that triggered new ideas and alternative insights. Thus, as organizations enter an era of transition (Tell et al., 2020), we find the term emancipation to be applicable to reveal current challenges for people and organizations. The emancipation perspective is mainly found in the category *create readiness to change and deal with paradoxes,* with unpredictable and constant change as a shared backdrop. For instance, principle (P14), coping with uncertainty by using concepts from strategic

Table 6. Knowledge Perspectives in the Principles

Principles	Technical Perspective	Understanding Perspective	Emancipatory Perspective
Balance the portfolio			
(P1) Find priority criteria	X		
(P2) Select the right projects	X		
(P3) Use tools to support decision-making	X		
(P4) Managing portfolios in multiple project environments	X		
(P5) Comprehend the deficiencies of computed methods	X		
Build a coherent PPM system			
(P6) Increase maturity by adopting best practices	X		
(P7) Apply a process view	X		
(P8) Focus on governance	X		
(P9) Secure a supportive culture		X	
Secure strategy effectiveness and stakeholder involvement			
(P10) Focus on strategic effectiveness		X	
(P11) Apply a holistic value approach		X	(X)
(P12) Focus on stakeholder involvement, roles, and responsibilities		X	
(P13) Use alternative understandings		X	
Create readiness to change and deal with paradoxes			
(P14) Cope with uncertainty			X
(P15) Establish preconditions for innovation and change			X
(P16) Enable agility outside of individual projects			X
(P17) Deal with paradoxes in transitional times			X

management theory, can be used to foster deliberate organizational changes. These categories' principles help provide optimistic outlooks and ways to cope with an unpredictable future. However, all the principles associated with the emancipatory perspective lose momentum when providing a critical view of the potentially damaging effects of project portfolios in local and global environments. Notwithstanding, if we look at the principles in the understanding perspective category, we find that principle (P11) provides a fruitful path to becoming more aware of ethical responsibilities and developing value orientation toward a more comprehensive set of responsibilities (Sankaran et al., 2021). This principle applies a holistic value approach, which includes a broad conceptualization of value (Laursen & Svejvig, 2016). This opens a path to see value as going beyond the organization's profit objective and its likelihood of success (Martinsuo & Killen, 2014), including social and environmental sustainability (Brook & Pagnanelli, 2014).

Despite some progress in this area, we argue there is a potential to strengthen the emancipatory perspective in PPM principles. To address this potential, we draw on Geraldi and Söderlund's (2016) arguments suggesting that emancipatory endeavors can be formed by so-called *engaged practitioners*—understood as informed and ethically acting practitioners.

Engaged Practice, Engaged Scholarship, and Emancipatory Principles

We argue that research is crucial in providing principles with an emancipatory perspective. As research is meant to be independent, researchers are in a privileged position to ask questions and enter into informed and critical dialogue. In organizations, these conditions are not always the case. Thus, practice–collaborative research approaches, such as engaged scholarship (Van de Ven, 2007), design science (Hevner et al., 2004), or action research (Svejvig & Schlichter, 2020), seem promising. First, developing principles with an emancipatory perspective is mutually relevant for research and practice. Second, the use of practice–research collaboration will move us beyond the simplistic notion that research principles are generated in the research lab and then packaged and diffused into practice (Mathiassen & Nielsen, 2008). Third, researchers have a responsibility to help organizations in developing and following ethical principles. This is crucial as we enter an era where moral and environmental responsibilities are expected to escalate. Thus, we argue that the time has come for research—in collaboration with practice—to develop more principles with an emancipatory perspective to cope with the environmental and social challenges of the upcoming decades (Walker & Lloyd-Walker, 2019).

Conclusion and Implications

This article contributes to what we know about PPM principles and pursues the research questions: *(1) How can we understand the intentions and assumptions underlying PPM principles? (2) How can we lay out a path for the future?* Our systematic literature review addresses these questions by tracing the PPM literature back to its origin and it identifies 17 principles. We show how these principles are embedded in a semantical hierarchy that lays the groundwork for helping practitioners become knowledgeable agents in PPM-related situations. Principles do not dictate what to do in specific situations. Instead, they

more broadly tell us how to think in different situations and thus enable practitioners to act in unknown circumstances. Thus, raising awareness of what we know about PPM principles becomes more relevant than ever before as their guidelines help practitioners cope with future and unknown challenges.

Our study proposes a path for future research based on reflections on the orientation of the PPM principles and their underlying knowledge perspectives. Our research scope—from 1950 to 2020—limits our results by not including contributions thereafter. Also, our analysis focused solely on the 25 articles with the most Google Scholar (GS) citations within four categories of principles. These limitations may have influenced our results. Notwithstanding, we argue that our systematic and historical review reveals vital insights into what we know about PPM principles to meet future challenges.

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Appendix A. Search String. Adopted from Hansen and Svejvig (2022)

Databases	Search Strings	Criterion 1: Time Span Covers the Period 1950–2019	Criterion 2: Includes 10 Benchmark Papers*	Search Results
JSTOR	((Title: "Portfolio" OR Abstract: "Portfolio") AND "project") OR ("Portfolio selection" < 1960)	1950–1960	(1) Markowitz (1952)	89
Scopus (Search A)	Title: ("Portfolio") OR ("Multiproject") OR ("multi-project") AND (Abstract: "Portfolio" and "Project") OR (Abstract "Project studies")	–2019	(2) Cooper et al. (1997b) (3) (Archer & Ghasemzadeh, 1999) (4) Martinsuo and Lehtonen (2007) (5) Blichfeldt and Eskerod (2008)	988
Scopus (Search B)	(Title: "Portfolio approach to Information systems") OR (Title: "IT portfolio management")	–2019	(6) McFarlan (1981) (7) Jeffery and Leliveld (2004)	32
Science Direct	Title + Abstract + Keywords: ("Project portfolio management")	–2019	(3) Archer & Ghasemzadeh (1999) (4) Martinsuo and Lehtonen (2007) (5) Blichfeldt & Eskerod (2008)	467
IEEE Xplore database	(Document title: "Portfolio") OR Document title: "Multi project") OR Document title: "Multiproject") OR Document title: "Project selection")	–2019	(8) Baker and Pound (1964) (9) Schmidt and Freeland (1992)	209
Project Management Journal® database	(Document title: "portfolio")	–2019	(10) Müller et al. (2008)	147
Back-tracking and Google Scholar	Backtracking of references in Baker and Pound (1964)	1950–1964	(1) Markowitz (1952)	38
Total			The search results include all 10 articles from our test sample.	1,970

*We utilize the 10 benchmark papers found by Hansen and Svejvig (2022) and the research initiated by Hansen and Svejvig (2018).

Appendix B: Selection Criteria. Adopted from Hansen and Svejvig (2022)

Criteria	Reasoning and Examples for the Inclusion of Publications
Portfolio management of projects of all kinds	This concept is the focal point of our research. We broadly define PPM as the overall organizational ability to manage project portfolios strategically and holistically to better support the organization's success (Clegg et al., 2018; Killen & Hunt, 2010).
Model and technique applied to project portfolios	Most of the contributions conforming to this criterion are designed to support portfolio selection, and some of the present examples were included in the review (See Henriksen & Traynor, 1999).
Project selection processes	This concept includes the entire selection process. Publications utilizing this concept typically investigate how the selection of projects connects to other organizational areas and have primarily been published some decades ago (e.g., Pearson, 1974).
Project studies	The project studies' approach applies overarching definitions of various levels of analysis. About our selection of relevant publications, the focus of this concept varies, as do its implications for PPM (e.g., Söderlund, 2004).
Governance of projects	The concept of *governance of projects* is sometimes just defined as governance. Müller et al. (2014) describe the governance of projects as the collective project governance, thus referring to program- and portfolio-level issues. We also found use of the term PPM governance (e.g., Hansen et al., 2017).
PMO (project management office)	The term project management office (PMO) can be understood as the support office, often for a large project or the unit fulfilling a controlling, supporting, or coordinating role for one or more portfolios (Unger, Gemünden, et al., 2012).
Papers with explicit implications for PPM	Publications with direct implications for the PPM level were included. Our pool of publications also welcomes contributions that do not mention the word "portfolio" but still impact PPM (e.g., Laursen & Svejvig, 2016; Maniak & Midler, 2014; e.g., Midler et al., 2019).
Papers about PPM but using a different wording	This criterion is designed to capture the older publications before the discipline started to use the word portfolio, which dealt with PPM-related matters (e.g., Klein & Meckling, 1958).